

M-Lab Monthly Stats

What gets measured, how to read it, and how [you](#) can explore it — today.

A 5-minute walk-through of the dataset using Jupyter Notebooks

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Monthly Stats Data

Every month, **millions of people** run an open speed test against [M-Lab](#).

We take **every** one of those tests and statistically summarize them in a **small, pre-computed file you can easily download**.

The notebooks in this tutorial turn that file into answers about any country, region, city, or provider in the world.

Where the data come from

- **NDT** — the Network Diagnostic Tool — the key tool in the open speed test which M-Lab runs.
- **Millions** of tests every month, from real user devices worldwide.
- **BigQuery** stores the raw rows; the full archive is **petabytes** of measurements.

The **Monthly Stats** dataset is M-Lab's curated, summarized view of all that:

- **2009 → present**, every month
- **240 countries**, plus regions, cities, and providers
- **one small Parquet file per month, per geography** (a few MB each)

Instead of querying petabytes of raw rows, each month you download one small file with pre-computed percentile values — for each metric, geography, and month.

What's inside each file

Metric	Column prefix	Unit	Better
Download throughput	download_p*	Mbit/s	↑ higher
Upload throughput	upload_p*	Mbit/s	↑ higher
Latency (min RTT)	latency_p*	ms	↓ lower
Packet loss rate	loss_p*	fraction	↓ lower

Where *p** is the percentile one of `['p1', 'p5', 'p10', 'p25', 'p50', 'p75', 'p90', 'p95', 'p99']` allowing you to explore the distribution and its skew

Percentiles: line everyone up

Line all the tests in a month up, slowest → fastest:

- **p50** — the median, the test in the exact middle — the typical user's experience
- **p95** — the fastest connections (or lowest loss / latency)
- **p1** — near the very back (slowest, highest loss / latency)

This shows us the way the metric (throughput, latency, loss) varies across the tests.

A wide p50→p95 gap shows how different experiences can be in the aggregation area
(country, subdivision, city, ASN / ISP)

The splits: "one row per what" is the only difference

Slice	One row per...
downloads_by_country / uploads_by_country	country
downloads_by_country_subdivision1	state / province
downloads_by_country_city	city
downloads_by_country_asn	provider (ASN)

Every slice keeps the **same four metrics** across their percentiles — only the geography changes.

What the numbers look like — June 2026, country medians

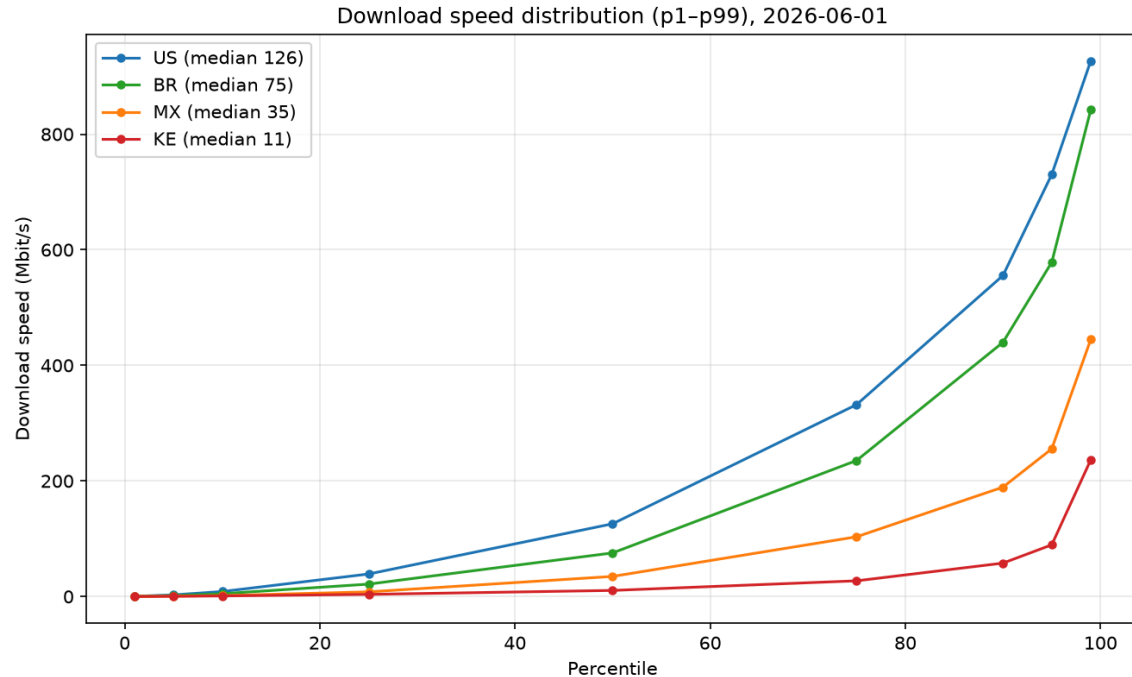
Download p50, Mbit/s (240 countries on file)

 US	 BR	 MX	 KE
126	75	35	11

What shape means here

- A **median** is one point — the typical user.
- The **distribution** is the whole curve: p5, p25, p50, p75, p95, p99.
- The notebook 01 **percentile sweep** shows whether a ranking survives at p95 vs p50 — often it does not.

The shape behind the medians — percentile curves



Each line is one country's **full distribution** (p1 → p99), June 2026. The **median** is just the middle point — the **curve is the story**: where it sits low and flat, that is where most users actually are; a stretch at the top is a minority running far faster. Notebook 01 draws these curves for any country, any metric, any month.

The four notebooks

	Question it answers
00 · introduction & catalog	What are these data, and how to load them?
01 · country explorer	How do I read a country's distribution shape?
02 · the splits	Where does it vary: countries, regions, cities, providers?
03 · multiple months	How do measurements change over months?

Each is **self-contained**

Try it right now — MyBinder

Click a notebook — it launches in your browser in seconds. No local install.

Notebook	Binder
00 · introduction & catalog	
01 · country explorer	
02 · the splits	
03 · multiple months	

Your country. Look up your own — median, shape, regions, and whether it improves.

Recap

- **Percentiles, not averages** — and the shape of the distribution can be used to ask many questions of the data.
- **Four metrics × the same slices** — country, region, city, Internet provider (ASN).
- **One loading pattern**, four notebooks, zero installs: launch any in MyBinder.

Thanks! Questions? Hands-on time — open a notebook and explore!